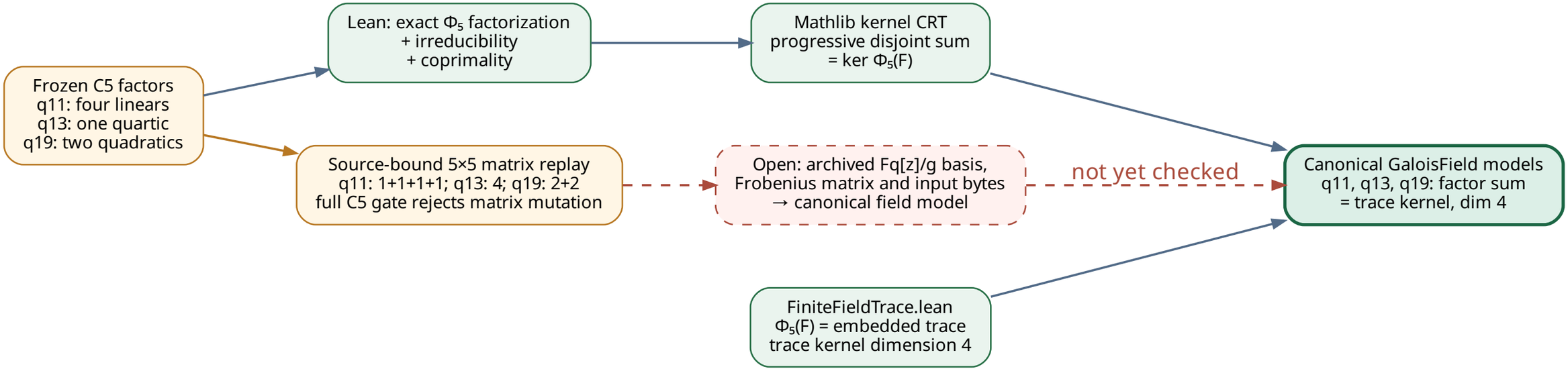
Frozen C5 factors
q11: four linears
q13: one quartic
q19: two quadratics
Lean: exact Φ5 factorization
+ irreducibility
+ coprimality
Mathlib kernel CRT
progressive disjoint sum
= ker Φ5(F)
Source-bound 5×5 matrix replay
q11: 1+1+1+1; q13: 4; q19: 2+2
full C5 gate rejects matrix mutation
Open: archived Fq[z]/g basis,
Frobenius matrix and input bytes
→ canonical field model
not yet checked
Canonical GaloisField models
q11, q13, q19: factor sum
= trace kernel, dim 4
FiniteFieldTrace.lean
Φ5(F) = embedded trace
trace kernel dimension 4